

Knight-distance labellings of a strip: OEIS A253337

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14 September 2026

Abstract

OEIS A253337 carries an empirical recurrence of order 49 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The entry's condition is decided by a bounded amount of state carried down the array; the admissible windows are the vertices of a finite digraph and the arrays counted are exactly the walks in it. Hence $a(n)$ is a walk count on $S = 1084$ vertices and is C -finite, and whether the conjectured recurrence annihilates it is settled by a finite exact computation, with the Cayley–Hamilton theorem bounding the work.

2020 Mathematics Subject Classification. 05A15, 05B45, 15B36.

1 The conjecture

OEIS A253337 is “Number of $(n+2) \times (3+2)$ nonnegative integer arrays with all values the knight distance from the upper left minus as much as 3, with successive minimum path knight move differences either 0 or +1, and any unreachable value zero.”. It has offset 1 and begins

1928, 11581, 100512, 722826, 5136108, 32653458, 237047904, 1348380871, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: $a(n) = a(n-1) + 12*a(n-4) - 12*a(n-5) - 66*a(n-8) + 66*a(n-9) + 220*a(n-12) - 220*a(n-13) - 495*a(n-16) + 495*a(n-17) + 792*a(n-20) - 792*a(n-21) - 924*a(n-24) + 924*a(n-25) + 792*a(n-28) - 792*a(n-29) - 495*a(n-32) + 495*a(n-33) + 220*a(n-36) - 220*a(n-37) - 66*a(n-40) + 66*a(n-41) + 12*a(n-44) - 12*a(n-45) - a(n-48) + a(n-49)$ for $n > 71$.

As of the “Last modified” line on the live entry (Sep 27 2015 12:20:57, revision 9) this is still recorded as empirical, and nothing on the entry records it as proved.

2 From values to a labelling

Write $d(c)$ for the knight distance from the top left corner of the board and $v(c)$ for the entry's value at c . The entry asks for $d(c) - 3 \leq v(c) \leq d(c)$, and for successive values along a minimum knight path to differ by 0 or +1. Put

$$w(c) = d(c) - v(c).$$

Then w takes values in $\{0, 1, \dots, 3\}$, and for an edge $a \rightarrow b$ of a minimum path — so $d(b) = d(a) + 1$ — the condition $v(b) - v(a) \in \{0, 1\}$ reads

$$v(b) - v(a) = (d(b) - d(a)) - (w(b) - w(a)) = 1 - (w(b) - w(a)) \in \{0, 1\} \iff w(b) - w(a) \in \{0, 1\}.$$

So the entry counts the labellings w that rise along every minimum-path knight move, by at most one each time. Cells the knight cannot reach carry the value zero and are not labelled; they contribute a factor of one.

3 The distance field of the strip

Everything below rests on two properties of d on the strip $S_5 = \{(i, j) : i \geq 0, 0 \leq j < 5\}$, and both are proved here rather than measured. Write d for the knight distance from $(0, 0)$ on S_5 and d_H for the knight distance on the board of exactly H rows.

Lemma 1. *Fix a width C and let $D : S_C \rightarrow \mathbb{N} \cup \{\infty\}$ be any function at all, satisfying (a) $D(0, 0) = 0$; (b) $D(y) \leq D(x) + 1$ for every knight-adjacent pair x, y ; and (c) every $x \neq (0, 0)$ with $D(x) < \infty$ has a knight neighbour y with $D(x) = D(y) + 1$. Then $D = d$.*

Proof. By (b) and induction on $d(x)$, $D(x) \leq d(x)$: the last step of a shortest path to x is an edge from a cell at distance $d(x) - 1$. Conversely (c) forces $D(x) \geq 1$ for $x \neq (0, 0)$, so $D(x) = 0$ only at the corner, and induction on $D(x)$ gives $d(x) \leq D(y) + 1 = D(x)$ with y the neighbour (c) supplies. Neither induction uses where D came from. $\square$

Proposition 2. $d(i + 4, j) = d(i, j) + 2$ for every j and every $i \geq 5$, and the unreachable cells repeat with the same period.

Proof. Take the field on rows $0, \dots, 12$ as computed by breadth-first search, and DEFINE D on the rest of the strip by $D(i + 4, j) = D(i, j) + 2$ for $i \geq 5$. Conditions (a), (b) and (c) of Lemma 1 at row i involve only rows $i - 2, \dots, i + 2$, because a knight move changes the row by 1 or 2. Once all five of those rows lie at or above 5, the definition gives $D(r, j) = D(r - 4, j) + 2$ for each of them, so the condition at row i is the condition at row $i - 4$ with 2 added to both sides, hence the same condition. Checking rows $0, \dots, 25$ therefore settles every row, and those checks were carried out in exact integer arithmetic. By Lemma 1, $D = d$, which is the claim. (The inequality $\leq$ needs no certificate: the moves $(2, 1)$ then $(2, -1)$ — or $(2, -1)$ then $(2, 1)$ at the right-hand edge — drop four rows and return to the same column. It is $\geq$ that the certificate supplies.) $\square$

Proposition 3. $d_H(i, j) = d(i, j)$ for every $H \geq 5$ and every $i < H$.

Proof. For a cell x let $\rho(x)$ be the least number of rows a board needs for x to sit at distance $d(x)$ on it, so $\rho(0, 0) = 1$ and, minimising over the minimum-path predecessors y of x ,

$$\rho(x) = \min_y \max(\text{row}(x) + 1, \text{row}(y) + 1, \rho(y)).$$

A board of H rows realises d on all of its own rows exactly when $H \geq \rho(x)$ for every x in them. By Proposition 2 the quantity $\rho(x) - \text{row}(x)$ has period 4 in the row index, so the condition is decided by finitely many rows; it holds for every $H \geq 5$ and fails at $H = 4$. $\square$

So one distance function serves every board of 5 rows or more. The heights 2, 3 and 4 are genuinely different — on a 3×3 board the centre is unreachable, while on any taller board it is at distance 4 — and are counted separately, each on its own board with its own distances, which needs no theorem at all.

4 The count is a walk count

What a transfer needs to know at row i is which pairs of cells within two rows are minimum-path edges, and how far each cell's label is capped by $\min(3, d)$. Both are functions of d on rows $i - 2, \dots, i$, so by Proposition 2 both depend on i only through $i \bmod 4$ once $i \geq 7$ — the cap because d rises with i and has passed 3 there.

A knight move never stays inside a row, so no edge joins two cells of one row: the label of a new row is constrained only by the two rows above it, and cell by cell at that. Take as vertices the pairs (labelling of the previous row, labelling of the current row) together with the phase

$i \bmod 4$ of the current row; an edge to the next such vertex exists when every minimum-path edge closed by the arriving row rises by 0 or 1, which the two rows above it decide. Vertices with identical futures are merged, which changes no count, leaving $S = 1084$.

Rows $0, \dots, 7$ are ABOVE the periodic region and are not carried by the matrix. They are walked once with weights: ι is the vector counting, at each vertex, the boards of 8 rows whose last two rows are that pair. So ι is not an incidence vector, and the paper may not call it one; τ is all ones, every vertex being a legitimate last pair. Carrying the row index in the vertex instead, which is the obvious alternative, multiplies the vertex set by the 7 distinct rows above the region and is what made the wider boards unreachable. Summing the early rows away loses nothing: a board is determined by its rows, and two boards agreeing on their last two rows admit the same continuations. With M the adjacency matrix, for every board of at least 8 rows,

$$a(n) = \iota^\top M^{n-6} \tau,$$

the exponent fixed by the entry's own shape and checked against its published terms. In particular a is C -finite.

5 The proof

Lemma 4. *Let $q(t) = t^{49} - \sum_i c_i t^{49-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j , so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j , and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+49} - \sum_i c_i M^{j+49-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M ; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. A constant factor in front of a divides out of the residual and changes nothing here. $\square$

Theorem 5.

$$a(n) = a(n-1) + 12a(n-4) - 12a(n-5) - 66a(n-8) + 66a(n-9) + 220a(n-12) - 220a(n-13) - 495a(n-16) + 495a(n-17)$$

for every $n > 71$.

Proof. The vector $w = q(M)\tau$ was formed by 49 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots$, again exactly; those integers vanish from the index corresponding to $n = 71 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. $\square$

6 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the reading was pinned before the digraph was written, by a program that enumerates every labelling of the board one by one, computing the knight distances by breadth-first search on that very board and testing each minimum-path edge directly. That program shares no code with the transfer matrix, so an error in one does not hide an error in the other. A third, independently written row-pair model, which carries the phase in the vertex rather than the row index and merges by a different rule, reproduces the same published terms again.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 4 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

The certificate of Propositions 2 and 3 was itself tested on cases with known answers before being trusted: the field it produces agrees cell for cell with a direct breadth-first search on every board of width 3 to 9 and every height up to 59; the threshold it computes is tight, the height below it genuinely disagreeing; and perturbing a single entry of its table by one makes it refuse.

References

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